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Dating mafic–ultramafic intrusions by ion-microprobing contact-melt zircon: examples from SW Sweden

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Abstract Zircons from anatectic melts of the country rocks of three Proterozoic mafic–ultramafic intrusions from the Sveconorwegian Province in SW Sweden were microanalyzed for U–Th–Pb and rare earth elements. Melting and interaction of the wall rocks with the intrusions gave rise to new magmas that crystallized zircon as new grains and overgrowths on xenocrysts. The ages of the intrusions can be determined by dating this newly crystallized zircon. The method is applied to three intrusions that present different degrees of complexity, related to age differences between intrusion and country rocks, and the effects of post-intrusive metamorphism. By careful study of cathodoluminescent images and selection of ion probe spots in zircon grains, we show that this approach is a powerful tool for obtaining accurate and precise ages. In the contact melts around the 916 ± 11 Ma Hakefjorden Complex, Pb-loss occurred in some U-rich parts of xenocrystic zircon due to the heat from the intrusion. In back-veins of the 1624 ± 6 Ma Olstorp intrusion we succeeded in geochemically distinguishing new magmatic from xenocrystic zircon despite small age differences. At Borås the mafic intrusion mixed with country rock granite to form a tonalite in which new zircon grew at 1674 ± 8 Ma. Reworking of zircon occurred during 930 + 33/–34 Ma upper amphibolite facies Sveconorwegian metamorphism. Pb-loss was the result of re-equilibration with metamorphic fluids. REE-profiles show consistent differences between xenocrystic, magmatic, and metamorphic zircon in all cases. They typically differ in Lu/La_N, Ce/Ce*, and Eu/Eu*, and igneous zircon with marked positive Ce/Ce* and negative Eu/Eu* lost its anomalies during metamorphism.

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Introduction

Precise dating of mafic-ultramafic intrusions is difficult, especially in polymetamorphic regions. Rb–Sr, Sm–Nd, and Ar dating commonly produce metamorphic or poorly constrained igneous ages. Mafic rocks contain little magmatic zircon and/or baddeleyite. Furthermore baddeleyite can easily be converted to zircon during metamorphism (e.g., Wingate et al. 1998).

Here we present a new approach to dating mafic–ultramafic intrusions using high resolution ion microprobes (SHRIMP and Cameca ims 1270 at the NORDSIM facility) for U–Pb dating of zircon that grew in back-veins or mixed rocks originating from partially melted country rock. In addition to dating the intrusions, this approach provides geochronological data for the country rocks and valuable insights into the behaviour of xenocrystic zircon in contact metamorphic and magmatic environments.

Three intrusions in SW Sweden were selected as test cases and represent examples of progressively greater complexity (Fig. 1). They are the Hakefjorden Complex, Olstorp intrusion, and Borås intrusion shown in Fig. 2. The Olstorp and Borås intrusions experienced later metamorphism at mid and upper amphibolite facies respectively. At each intrusion contact metamorphism produced anatectic melts from which new zircon formed.

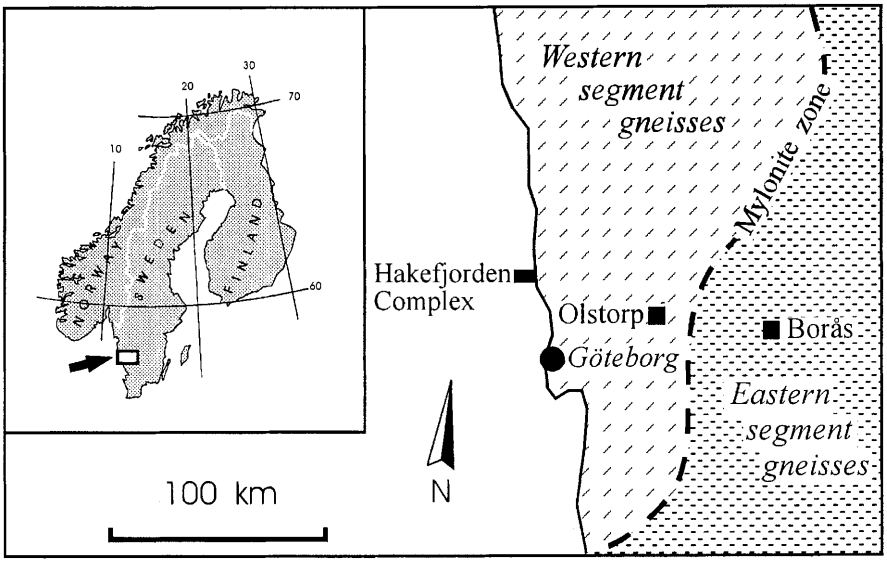


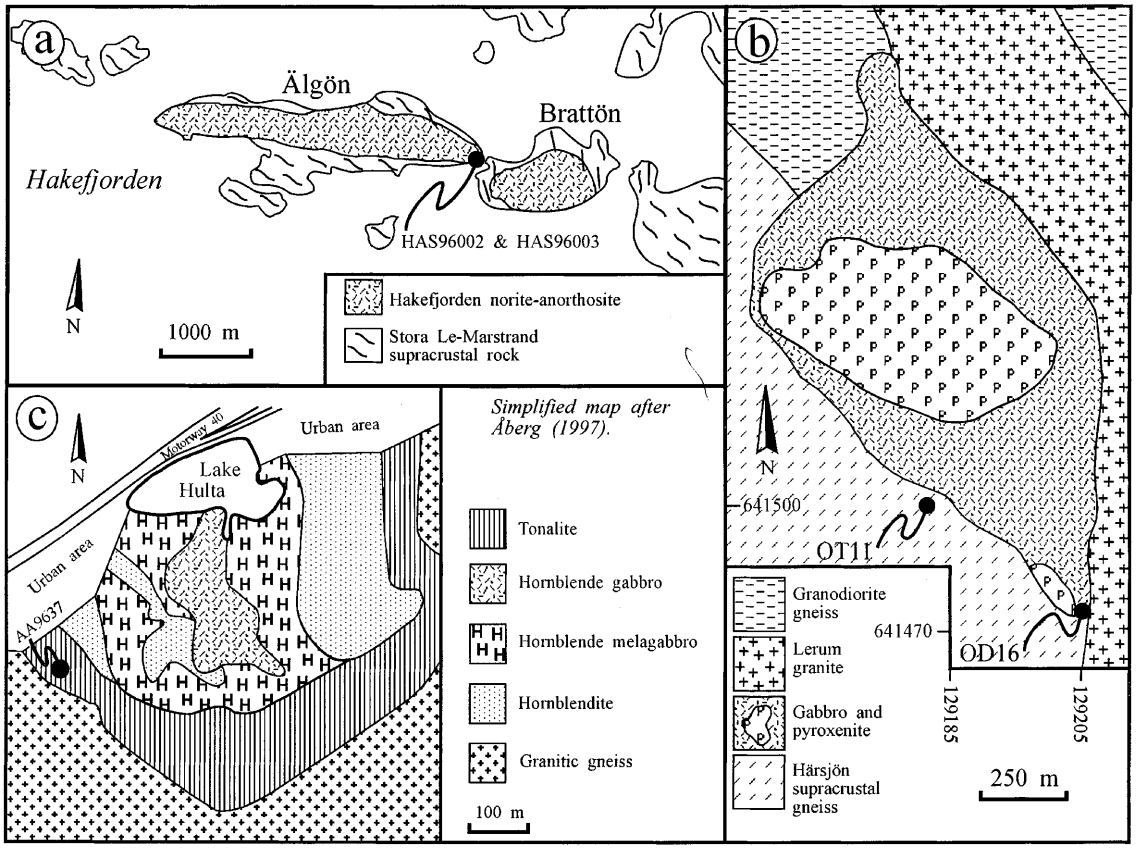
Fig. 1 General geology of SW Sweden. The locations of Hakefjorden Complex, Olstorp intrusion and Borås intrusion are indicated by black boxes

Field characteristics and samples

The noritic Hakefjorden Complex intruded >1587 ±3 Ma Stora Le-Marstrand metagreywackes (Connelly and Åhäll 1996). Årebäck (1995) reported a 959 ± 74 Sm–Nd mineral isochron for the norite. Sample HAS96002 was collected from a 5–20 m wide anatectic migmatite that developed along the contact (Årebäck 1995). HAS96003 was taken from a small granitic dyke which intrudes the Hakefjorden Complex from the migmatite as a back-vein (Fig. 3a).

The Olstorp intrusion comprises well-preserved ultramafic olivine–hornblende–pyroxenite cumulates in unaltered hornblende gabbro and amphibolite facies metagabbro. A 5-m long profile

Fig. 2 Simplified geological maps for **a** Hakefjorden Complex, **b** Olstorp intrusion and **c** Borås intrusion. Sample locations are indicated with sample number. The Olstorp map is partly based on gravimetry and magnetometry and coordinates are given in the Swedish national grid



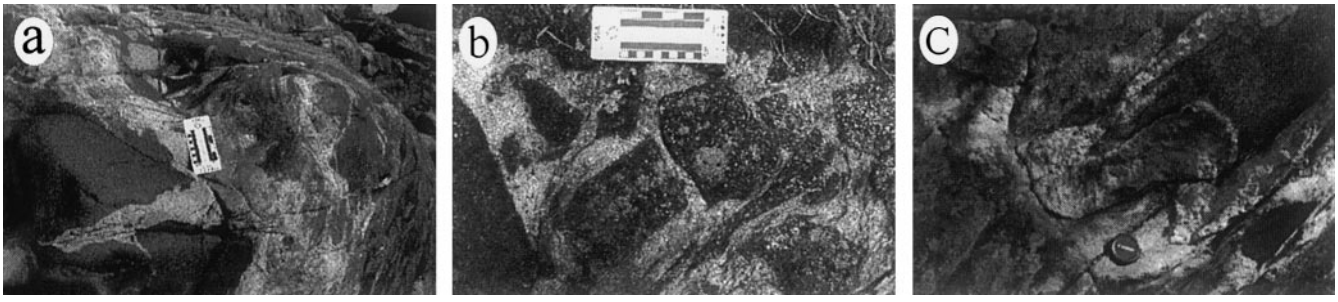


Fig. 3 Contact relationships and sample location for **a** Hakefjorden Complex, **b** Olstorp intrusion and **c** Borås intrusion. At Hakefjorden Complex the intruded sediment is increasingly affected by the intruding norite-anorthosite magma. Partial melting of the sediment result in granite veins that back intrude the norite–anorthosite body. At Olstorp granite veins form a net-like pattern with interlacing rounded gabbro blocks. The Borås picture shows the foliated contact tonalite with folded granitic veins of irregular thickness and orientation. All were taken at or next to sample localities

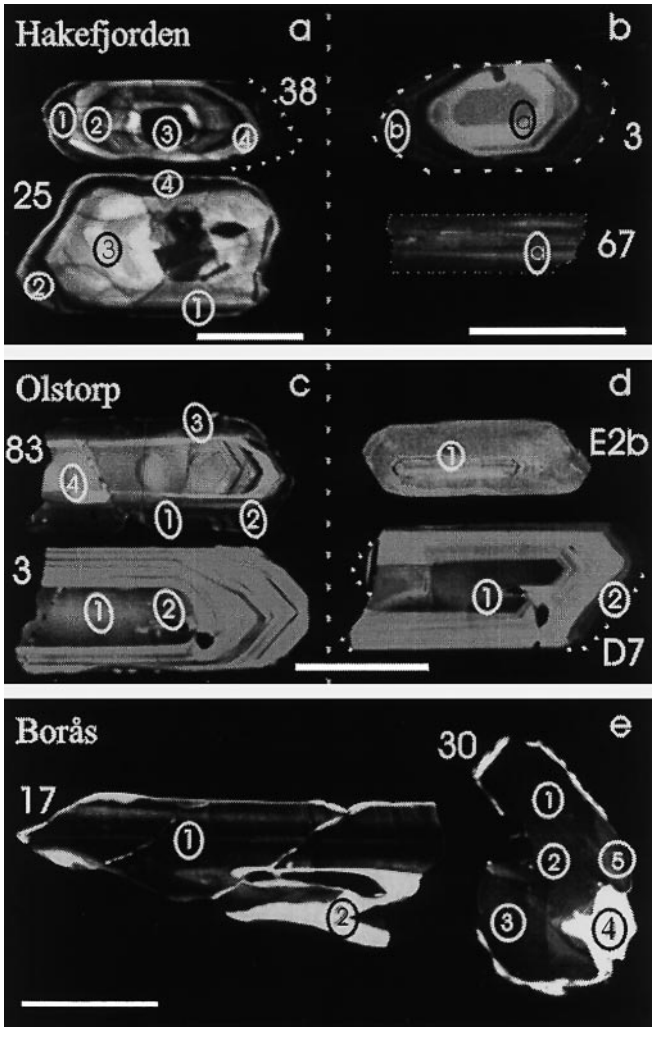


Fig. 4a–e Cathodoluminescence images of selected zircons from each sample. **a** HAS96002, **b** HAS96003, **c** OT11, **d** OD16 and **e** AA9637. Grain numbers are indicated next to each grain and with spot locations shown with spot numbers. All scale bars are 100 μm

from relatively undisturbed paragneiss into the net vein complex occurs at the southern end of the intrusion. The net vein complex is characterized by rounded and elongated blobs of gabbro in a granitoid matrix that sometimes contains dark smeared-out gabbro bands (Fig. 3b). The granitoid vein sample OD-16 was collected here. Sample OT-11 of the country rock paragneiss was taken just west of the intrusion as shown in Fig. 2b.

The Borås intrusion consists of hornblendite, hornblende melagabbro, hornblende gabbro and a foliated tonalite that surrounds the intrusion (Fig. 2c). Field observations and geochemical data of Åberg (1997) support a fractional crystallization model for the hornblendite to hornblende gabbro, but the tonalite sampled as AA9637 was produced by mixing between the country rock and a mafic component. Later Sveconorwegian metamorphism at Borås reached upper amphibolite facies at 9 kbar and 700 °C, leading to the development of unfoliated garnet–hornblende–plagioclase–biotite parageneses in some parts of the gabbro (Åberg 1997).

Methods

Zircon preparation

Zircons were separated by crushing, sieving and panning or using heavy liquids. To avoid contamination a Wilfley table was not used. Zircons were mounted in epoxy resin, polished (mainly exposing the outer zones) and gold-coated for cathodoluminescence (CL) studies and ion probe dating.

The zircons were examined using a Zeiss DMS 940 scanning electron microscope at 25 kV accelerating voltage. The monochromatic cathodoluminescence detector was coupled to a Oxford LINK EDS system to obtain digital images. The CL-emission is not calibrated against a standard, so "dark" or "bright" CL-images are relative to the resin and to other grains in the mount. Spots were selected using CL-images as shown in Fig. 4.

Analytical techniques

Secondary ion mass spectrometry (SIMS) dating was done using 15–35 μm spots as described by Compston et al. (1984) for SHRIMP II and Whitehouse et al. (1997, 1999) for Cameca ims 1270 (NORDSIM) respectively. Concentrations of P, Y, REE, Hf, Th and U in zircons were determined by SIMS using the SHRIMP I as described by Hoskin (1998). In brief, an O_2^- primary-ion beam is focused onto the zircon and the secondary ions produced are analyzed in a high-resolution mass spectrometer after passing through a series of ion lenses and steering plates.

Some zircon grains were also analyzed using a Cetac LSX200 UV laser coupled to a HP4500 ICP mass spectrometer. For each spot, 20 seconds of background was collected before the 50 μm laser beam was activated with a 4 Hz pulse rate vaporizing the zircon for 80 seconds with a penetration rate of 0.1 μm per shot. ^{30}Si was used as the internal standard and NIST 612 glass was used for calibration after every five analyses. The time-resolved spectra for several elements were inspected for indications of contamination (e.g., a K peak early in the analysis), inclusions or changes in

zircon composition. One background and three peak time intervals each containing data for about 20 seconds were selected and reduced using a spreadsheet. Analyses of zircon standard 91500 suggests accuracy around 5% for REE which is comparable with that of the ion-probe.

The program ISOPLOT by Ludwig (1991, 1998) was used for all age calculations. Unless stated, errors are all given at the 2σ-level of confidence. In order to minimize problems related to multiple lead loss events, our preferred mode of calculation is the concordant three ratio ($^{206}Pb/^{238}U$, $^{207}Pb/^{235}U$ and $^{207}Pb/^{206}Pb$) age calculation described by Ludwig (1998) as *concordia ages* and implemented in version 2.96 and *Ex*1.00b of ISOPLOT, with MSWD as a measure of concordance and equivalence of the group. The age calculations include decay constant errors. Discordant analyses were regressed in Wetherill concordia diagrams to determine discordia intercepts. For regressions with essentially zero-age lower intercepts, the recent lead loss model is assumed to be valid and $^{207}Pb/^{206}Pb$ ages are used. U–Pb results are presented in Table 1, age determinations for individual grains or groups in Table 2, and REE-data in Table 3.

Results

Hakefjorden Complex: contact-melt zircon much younger than xenocrysts

HAS 96002 from the Hakefjorden Complex contact migmatite only contains inherited zircon from the Stora Le-Marsrand metagreywackes with some thin CL-dark rims as shown in Fig. 4a. Zircon from HAS96003 granite back-vein is also dominated by xenocrystic zircon. However, six broken needle-like grains with CL-dark images and faint oscillatory zoning, illustrated in Fig. 4b, form a distinct morphological group that is thought to represent zircon grown in the anatectic back-vein. Laser ablation ICPMS REE analyses of zircon plot in two groups. The xenocrysts have Lu/La_N-ratios between 34 and 220 and small negative Eu anomalies. The second group, of needle-like zircon has much higher Lu/La_N-ratios between 1021 and 5414 and more prominent Eu-anomalies (Fig. 6a and Table 3). This supports the identification of the needle-like zircon as a genetically distinct group. Age determinations of the xenocrysts from both contact migmatite and back-vein samples have upper intercept or concordia ages that overlap with or are older than the 1587 ± 3 Ma minimum age of the Stora Le-Marstrand metagreywackes. However individual grains show different lead loss characteristics as discussed later (Table 2). CL-dark overgrowths that sometimes cross-cut oscillatory zoning of detrital grains are interpreted as new zircon that grew during the anatectic event. That these rims are younger is indicated by their discordant ages but they are too thin to be resolved from the cores. The six needle-like grains from the back-vein cluster around 900 Ma on the concordia plot (Fig. 5c). Some degree of reverse discordance is observed but four spots are concordant within 3% (excluding error overlap) and give a concordia age of 916 ± 11 Ma (MSWD = 0.94). This is consistent with the Sm–Nd age, and is interpreted as the age of back-vein crystallisation soon after intrusion of the Hakefjorden Complex (Table 2).

Olstorp intrusion: contact-melt zircon with similar age to xenocrysts

All five grains investigated in the Härsjön paragneiss OT-11, intruded by the Olstorp intrusion, show oscillatory zoning in CL as illustrated in Fig. 4c. This paragneiss might have a heterogeneous source, so two to four spots were analysed in each of the five zircon grains. The analyses are plotted in Fig. 5d and concordia ages were calculated for all except one discordant grain, for which a $^{207}Pb/^{206}Pb$ age was calculated (Table 2). The ages vary from 1639 ± 22 to 1675 ± 17 Ma which should represent provenance ages. Th and U concentrations vary both between grains and within two of the grains. Neither Th and U concentrations nor Th/U ratios show any correlation with age for the different grains. The back-vein sample contains zircon with distinct morphological and chemical differences. None of the grains are concordant but they appear to plot along a recent lead loss line and we therefore use the $^{207}Pb/^{206}Pb$ ages in our calculations. Three groups of zircon can be identified based on crystal morphology, REE and Th–U chemistry, CL-images, and ages. The first group contains only one grain (D6) with a low aspect ratio, high U and Th, dark CL-image, relatively flat REE-profile (Fig. 6b, Table 3), and a $^{207}Pb/^{206}Pb$ age of 1664 ± 8 Ma. The second group contains rather stubby crystals, with lower U and Th and bright CL-images. Their REE-profiles have steep $Lu/La_N \geq 5600$, with large positive Ce-anomalies and negative Eu-anomalies. They show a range in $^{207}Pb/^{206}Pb$ ages between 1614 ± 44 Ma and 1665 ± 54 Ma which is similar to that found in the Härsjön paragneiss. The third group has higher aspect ratios than the other groups, with high U and Th, dark CL-images and REE-profiles lower than but similar to that of the first group (Figs. 4d and 6b). An overgrowth on a broken grain (Fig. 4d) is attributed to the third group based on similar REE-profiles, Th and U chemistry, and CL-image. This is an important observation since it indicates that group three is younger than group two. The group one grain with similar Th, U, and REE-chemistry is morphologically distinct and significantly older than group three zircons. We interpret the group three zircon to have formed during the back-veining event since this type of grain is absent in OT-11. The 1620 Ma regression in Table 2 supports the recent lead loss model which allows the 1624 ± 6 Ma (MSWD = 0.16) $^{207}Pb/^{206}Pb$ age to be used. By analogy with the Hakefjorden Complex, this age is interpreted to date the anatectic back-vein and thereby the Olstorp intrusion. The age overlaps with some of the xenocrystic grains but the combination of CL images, REE-profiles and relative ages of rim and core in grain D7 (Fig. 4d) support the separation between xenocrysts and magmatic grains.

Table 1 U–Pb zircon data for the Hakefjorden Complex, Olstorp and Borås samples. Xenocrystic or new zircon is indicated for the Hakefjorden Complex and Olstorp back-vein grains with x and n suffixes respectively. Sveconorwegian metamorphic zircon is indicated with an m suffix for the Borås tonalite. NORDSIM data is indicated by an n prefix. Sample localities are given according to the Swedish national grid

Sample Spot #	U ppm	Pb ppm	Th ppm	Th/U Meas.	$^{206}Pb/^{204}Pb$	$^{207}Pb/^{206}Pb$	±σ %	$^{206}Pb/^{238}U$	±s %	$^{207}Pb/^{235}U$	±σ %	ρ	Conc %	$^{207}Pb/^{206}Pb$	±σ	$^{206}Pb/^{238}U$	±σ	$^{207}Pb/^{235}U$	±σ
HAS 96002 Hakefjorden Complex contact migmatite. Coordinates: 125821/642930																			
n-A55a	235	91	57	0.24	30530	0.1102	0.43	0.3273	1.50	4.971	1.56	0.96	101	1802	8	1825	24	1814	13
n-A75a	706	155	319	0.45	950	0.0885	0.89	0.1718	2.56	2.096	2.71	0.94	73	1394	17	1022	24	1148	18
n-B40a	763	212	76	0.10	4390	0.1025	0.32	0.2429	1.15	3.432	1.19	0.96	84	1669	6	1402	14	1512	9
17.1	1050	322	131	0.12	20000	0.1081	0.46	0.3119	1.31	4.650	1.44	0.95	99	1768	9	1750	23	1758	25
20.1	846	266	782	0.92	183	0.0974	2.67	0.2649	1.36	3.558	3.15	0.54	96	1576	51	1515	21	1540	48
20.2	544	161	430	0.79	81	0.1181	5.33	0.2364	1.57	3.851	5.71	0.37	71	1928	98	1368	21	1604	92
24.1	2780	242	643	0.23	207	0.0879	5.92	0.0830	1.33	1.006	6.26	0.36	37	1380	118	514	7	707	44
24.2	919	205	253	0.28	5587	0.0988	0.71	0.2159	1.39	2.942	1.67	0.91	79	1602	14	1260	18	1393	23
25.1	815	129	388	0.48	3311	0.0847	1.06	0.1500	1.40	1.751	1.83	0.82	69	1308	21	901	13	1028	19
25.2	1393	164	457	0.33	658	0.0898	3.56	0.1163	1.38	1.440	4.03	0.49	50	1421	71	709	10	906	36
25.3	263	82	151	0.58	2639	0.1031	2.04	0.2810	1.78	3.993	2.88	0.71	95	1680	38	1596	28	1633	47
25.4	1944	242	569	0.29	354	0.0764	4.97	0.1211	6.77	1.275	8.94	0.83	67	1105	103	737	50	835	75
38.1	1635	333	170	0.10	954	0.0955	3.87	0.2110	1.94	2.778	4.50	0.52	80	1537	74	1234	24	1350	61
38.2	800	168	203	0.25	144	0.0892	5.04	0.2075	1.54	2.552	5.45	0.39	86	1408	99	1216	19	1287	70
38.3	2659	730	960	0.36	4673	0.1098	0.27	0.2614	1.22	3.955	1.29	0.98	83	1795	5	1497	18	1625	21
38.4	882	188	121	0.14	669	0.0997	1.50	0.2170	1.34	2.985	2.18	0.73	78	1619	29	1266	17	1404	31
42.1	283	92	240	0.85	2227	0.0988	1.82	0.2793	1.65	3.804	2.58	0.71	99	1601	34	1588	26	1594	41
42.2	376	120	228	0.61	100000	0.0996	0.50	0.2918	1.44	4.007	1.57	0.95	102	1616	8	1651	24	1636	26
42.2b	368	113	223	0.61	27027	0.0994	0.50	0.2790	1.40	3.824	1.52	0.94	98	1613	8	1586	22	1598	24
42.3	1482	340	332	0.22	1479	0.0853	1.41	0.2250	2.04	2.648	2.61	0.84	99	1323	27	1308	27	1314	34
42.4	577	173	453	0.78	549	0.1033	2.81	0.2584	1.43	3.682	3.34	0.56	88	1685	53	1482	21	1568	52
HAS 96003; Hakefjorden Complex granite back-vein. Coordinates: 125821/642930																			
n-03ax	78	33	79	1.01	59880	0.0995	0.66	0.2971	2.17	4.076	2.27	0.96	104	1615	12	1677	32	1649	18
n-03bx	402	146	201	0.50	15150	0.0976	0.39	0.2932	1.37	3.945	1.42	0.96	105	1579	7	1657	20	1623	11
n-20ax	704	216	169	0.24	98910	0.0956	0.39	0.2620	1.40	3.455	1.46	0.96	97	1541	7	1500	19	1517	11
n-21ax	152	59	145	0.95	31450	0.0972	0.63	0.2834	2.51	3.796	2.59	0.97	102	1570	12	1608	36	1592	21
n-21bx	2714	304	1408	0.52	2410	0.0795	0.98	0.1005	1.33	1.102	1.65	0.81	52	1185	19	617	8	754	9
n-28an	250	56	220	0.88	4100	0.0685	0.77	0.1667	1.53	1.575	1.72	0.89	112	884	16	994	14	960	11
n-28bn	238	48	212	0.89	6200	0.0689	0.87	0.1509	1.68	1.433	1.89	0.89	101	895	18	906	14	903	11
n-28cn	249	54	196	0.79	26540	0.0668	0.68	0.1653	1.82	1.522	1.94	0.94	119	831	14	986	17	939	12
n-28dn	231	48	179	0.77	16490	0.0672	0.66	0.1603	1.65	1.484	1.78	0.93	114	843	14	959	15	924	11
n-28en	158	31	116	0.73	116750	0.0701	1.03	0.1534	2.06	1.483	2.31	0.89	99	932	21	920	18	923	14
n-38ax	730	230	1176	1.61	580	0.0932	0.89	0.2079	1.65	2.673	1.88	0.88	82	1493	17	1218	18	1321	14
n-41ax	2014	265	143	0.07	320	0.0663	4.44	0.1057	1.34	0.967	4.64	0.29	79	817	93	647	8	687	23
n-41bx	300	25	112	0.37	160	0.0832	3.97	0.0535	2.43	0.614	4.65	0.52	26	1275	77	336	8	486	18
n-45ax	699	132	628	0.90	990	0.0929	1.70	0.1321	1.94	1.692	2.58	0.75	54	1486	32	800	15	1006	16
n-45bx	391	77	842	2.16	480	0.0911	1.26	0.1284	2.86	1.614	3.12	0.92	54	1450	24	779	21	975	19
n-50an	326	69	235	0.72	7910	0.0692	0.56	0.1650	1.87	1.574	1.96	0.96	109	905	12	984	17	960	12
n-50bn	271	55	220	0.81	5310	0.0681	1.00	0.1564	1.98	1.468	2.22	0.89	108	871	21	937	17	917	13
n-51an	399	82	380	0.95	47870	0.0693	0.48	0.1531	1.40	1.463	1.49	0.95	101	907	10	918	12	915	9
n-51bn	332	51	419	1.26	950	0.0688	1.26	0.1151	1.53	1.091	1.99	0.77	79	892	26	702	10	749	10
n-51cn	362	77	370	1.02	2250	0.0682	1.33	0.1588	1.50	1.493	2.01	0.75	109	874	28	950	13	928	12
n-57an	239	46	98	0.41	3340	0.0694	1.15	0.1573	1.61	1.505	1.98	0.81	103	910	24	942	14	932	12
n-57bn	426	57	140	0.33	530	0.0624	1.83	0.1045	1.76	0.899	2.53	0.69	93	688	39	641	11	651	12
n-61an	248	44	182	0.74	1030	0.0668	1.54	0.1428	1.76	1.315	2.34	0.75	104	831	32	860	14	852	13
n-61bn	326	48	304	0.93	400	0.0572	7.10	0.1109	2.88	0.875	7.66	0.38	136	500	157	678	19	638	36
n-62an	218	82	186	0.85	14120	0.0985	0.42	0.2771	1.57	3.764	1.63	0.97	99	1596	8	1577	22	1585	13
n-64ax	279	98	315	1.13	11970	0.0961	0.40	0.2433	1.59	3.223	1.64	0.97	91	1549	8	1404	20	1463	13
n-67an	292	51	444	1.52	390	0.0660	11.26	0.1274	2.71	1.159	11.58	0.23	96	806	237	773	20	782	61
OT 11; Olstorp intrusion country rock paragneiss. Coordinates: 129180/641500																			
11_3.3	233	76	194	0.83	1000000	0.1010	0.79	0.2816	2.91	3.923	3.11	0.97	97	1643	15	1599	47	1618	50
11_3.4	202	67	127	0.63	100000	0.1004	0.90	0.2987	2.78	4.136	3.00	0.95	103	1632	17	1685	47	1661	50
11_66.1	235	76	177	0.75	45455	0.1013	0.99	0.2822	2.73	3.942	2.99	0.94	97	1648	18	1603	44	1622	49
11_66.2	211	66	159	0.76	1000000	0.1032	0.78	0.2748	2.58	3.909	2.79	0.96	93	1682	14	1565	40	1616	45
11_66.3	169	52	122	0.72	500000	0.1033	0.77	0.2703	2.74	3.848	2.96	0.97	92	1684	15	1542	42	1603	47
11_83.1	1072	340	539	0.50	45455	0.1019	0.69	0.2968	2.56	4.170	2.71	0.97	101	1659	12	1676	43	1668	45
11_83.2	1311	425	647	0.49	22222	0.1042	1.15	0.3077	3.02	4.419	3.37	0.94	102	1700	21	1729	52	1716	58
11_83.3	1026	326	491	0.48	62500	0.1015	0.59	0.2972	2.59	4.160	2.72	0.98	102	1652	11	1678	43	1666	45
11_83.4	216	79	125	0.58	100000	0.1031	2.33	0.3373	16.07	4.795	16.54	0.99	112	1681	44	1874	301	1784	295
11_84.1	330	106	275	0.83	100000	0.1000	0.70	0.2815	5.75	3.881	5.87	0.99	99	1624	13	1599	92	1610	95
11_84.2	325	108	250	0.76	52632	0.0999	2.80	0.2892	6.12	3.983	7.03	0.92	101	1622	53	1637	100	1631	115
11_84.3	1057	317	512	0.48	250000	0.1020	0.69	0.2804	2.39	3.945	2.59	0.96	96	1661	13	1594	38	1623	42

Table 1 (Contd.)

Sample Spot #	U ppm	Pb ppm	Th ppm	Th/U Meas.	$^{206}Pb/^{204}Pb$	$^{207}Pb/^{206}Pb$	±σ %	$^{206}Pb/^{238}U$	±s %	$^{207}Pb/^{235}U$	±σ %	ρ	Conc %	$^{207}Pb/^{206}Pb$	±σ	$^{206}Pb/^{238}U$	±σ	$^{207}Pb/^{235}U$	±σ
11_85.1	945	299	552	0.58	20833	0.1028	0.58	0.2892	2.35	4.100	2.46	0.97	98	1676	10	1638	39	1654	41
11_85.2	1394	408	484	0.31	142857	0.1015	0.39	0.2851	2.31	3.991	2.38	0.99	98	1652	7	1617	37	1632	39
11_85.3	3490	1054	1341	0.38	166667	0.1031	0.29	0.2880	2.36	4.095	2.32	0.99	97	1681	5	1632	39	1553	36
OD 16; Olstorp intrusion net-vein granite. Coordinates: 129205/641472																			
16_d3bx	76	24	55	0.72	1988	0.0996	1.10	0.2748	2.37	3.773	2.76	0.92	92	1616	21	1523	36	1583	44
16_d4b1x	53	16	33	0.62	2941	0.1022	1.47	0.2679	3.10	3.776	3.60	0.91	84	1665	27	1367	42	1475	53
16_d6.1x	927	259	376	0.41	13158	0.1022	0.29	0.2665	2.03	3.754	2.08	0.99	102	1664	4	1645	33	1634	34
16_d7.1x	256	75	270	1.05	6944	0.1005	3.18	0.2362	4.40	3.273	5.77	0.84	95	1633	61	1547	68	1579	91
16_d7.2n	479	140	98	0.20	12500	0.0998	0.50	0.2908	2.13	4.000	2.25	0.98	97	1620	9	1586	34	1607	36
16_e2b1n	1276	354	494	0.39	4762	0.1000	0.20	0.2657	2.07	3.663	2.10	1.00	94	1623	4	1519	31	1563	33
16_e3ax	53	18	50	0.94	1988	0.1009	1.78	0.2884	2.74	4.010	3.42	0.85	93	1640	33	1520	42	1572	54
16_e3b1x	86	28	95	1.10	4049	0.1009	0.99	0.2660	2.89	3.701	3.16	0.95	95	1641	19	1533	44	1568	50
16_e6a1n	1048	301	511	0.49	4202	0.0999	0.40	0.2712	2.25	3.736	2.33	0.99	93	1623	7	1506	34	1557	36
16_e6bx	138	47	150	1.09	4115	0.1006	0.80	0.2790	2.29	3.868	2.51	0.95	100	1635	15	1634	37	1636	41
16_e7cx	77	25	77	1.00	2571	0.0995	1.21	0.2685	3.13	3.683	3.48	0.94	92	1614	22	1530	48	1588	55
16_e7bn	900	263	614	0.68	2227	0.1002	0.40	0.2631	2.09	3.633	2.14	0.98	97	1627	8	1565	33	1587	34
AA9637; Borås intrusion tonalite. Coordinates: 133080/640255																			
4.1	673	249	759	1.13	83333	0.1020	0.49	0.3003	2.70	4.223	2.60	0.98	102	1661	8	1693	46	1679	44
5.1	234	72	206	0.88	13333	0.1028	3.89	0.2526	4.80	3.581	2.45	0.59	87	1676	65	1452	70	1545	38
5.2	247	78	207	0.84	100000	0.0993	0.50	0.2681	1.44	3.671	1.31	0.94	95	1611	8	1531	22	1565	20
17.1	749	270	851	1.14	37037	0.1019	0.49	0.2917	1.54	4.097	1.37	0.95	99	1659	8	1650	25	1654	23
17.2	87	24	56	0.64	8197	0.0943	1.27	0.2416	2.55	3.140	2.03	0.87	92	1513	19	1395	36	1443	29
20.1	247	75	182	0.74	83333	0.0991	0.61	0.2680	1.61	3.662	1.42	0.93	95	1607	10	1531	25	1563	22
20.2	347	108	232	0.67	100000	0.0998	0.50	0.2776	1.92	3.181	1.44	1.00	98	1619	8	1580	30	1597	23
20.3	241	80	234	0.97	26316	0.0973	1.03	0.2790	1.90	3.744	1.51	0.84	101	1574	16	1586	30	1581	24
20.4	206	43	80	0.39	4219	0.0935	1.93	0.1933	2.65	2.493	1.60	0.69	76	1498	29	1139	30	1270	20
21.1	162	55	115	0.71	20833	0.1027	0.78	0.2991	2.27	4.234	2.01	0.94	101	1673	13	1687	38	1681	34
28.1	583	194	658	1.13	100000	0.1019	0.29	0.2669	2.00	3.751	1.91	0.99	92	1659	5	1525	30	1582	30
28.3	873	310	1050	1.20	100000	0.1014	1.08	0.2848	2.36	3.982	1.97	0.89	98	1650	18	1616	38	1630	32
30.1	943	328	842	0.89	66667	0.1028	0.29	0.2963	1.40	4.200	1.35	0.98	100	1676	5	1673	24	1674	23
30.2	567	196	548	0.97	90909	0.1017	0.59	0.2887	1.66	4.046	1.45	0.94	99	1655	10	1635	27	1644	24
30.3	648	210	378	0.58	38462	0.1030	0.68	0.2960	1.64	4.201	1.42	0.91	100	1678	11	1671	27	1674	24
30.4a* m	101	13	2	0.02	1642	0.0726	1.79	0.1416	5.28	1.325	2.40	1.20	99	863	15	854	45	857	21
30.4b* m	82	10	1	0.01	5495	0.0701	2.71	0.1347	4.40	1.319	2.08	0.89	85	957	26	815	36	854	18
30.4c m	54	11	7	0.13	333333	0.0868	1.27	0.1988	2.40	2.377	1.81	0.85	86	1355	17	1169	28	1236	22
30.5	1063	478	713	0.67	12821	0.1021	0.88	0.3987	1.91	5.612	1.58	0.89	130	1662	15	2163	41	1918	30
31.1* m	244	36	2	0.01	23810	0.0716	0.84	0.1607	1.80	1.607	1.43	0.89	96	1001	8	961	17	973	14
31.1b* m	191	30	1	0.01	100000	0.0728	0.96	0.1675	1.86	1.721	1.49	0.86	95	1056	10	998	19	1017	15
31.2	1454	526	1552	1.07	100000	0.1034	0.39	0.2964	1.40	4.226	1.28	0.96	99	1686	7	1674	23	1679	22
31.3	374	121	241	0.64	16667	0.1039	0.58	0.2917	1.56	4.178	1.37	0.93	97	1694	10	1650	26	1670	23

Borås intrusion: zircon from mixed magma, with metamorphic overprint

All zircon grains in AA9637 Borås tonalite are anhedral to subhedral, clear and variably fractured (Fig. 4e). No cores and only a few inclusions of other silicates were identified in transmitted light. Interference colours are very variable and some grains show well-developed zoning. The grains are mainly CL-dark with poorly developed zonation as seen in Fig. 4e. Zircons from the adjacent granite are very different being commonly euhedral with CL-bright cores and CL-dark metamorphic overgrowths. The tonalite grains are between 200 and 430 microns long, roughly twice the size of those from the granite and are interpreted as igneous grains that grew during crystallization of the tonalite. Some grains show subtle variations in CL-brightness, increasing towards CL-bright fracture fillings, embayments and rims (Fig. 4e). These cannot be identified optically and are interpreted as reworked parts of the igneous crystals. The CL-dark igneous grains show heavy REE enrichment, strong positive Ce-anomalies and negative Eu-anomalies (Fig. 6c; Table 3). One spot (17.2) that included both CL-dark and CL-bright material is essentially parallel but with lower concentrations than the CL-dark areas, retaining Ce- and Eu-anomalies. In contrast two CL-bright spots 30-A and 31-A have distinctly different REE-patterns with insignificant Ce-anomalies, modest Eu-anomalies, and lower heavy REE concentrations (Fig. 4e). Five concordant CL-dark spots give a concordia age of 1674 ± 8 Ma (MSWD = 0.80) (Fig. 5f, Table 2). This is interpreted to date the tonalite and thereby the Borås intrusion. A discordia of 1685 ± 16 Ma with a lower intercept of 930^{+33}_{-34} Ma (MSWD = 0.87) is obtained by adding CL-brighter discordant spots from the same grains that were used for the concordia age (Table 2). This indicates that the CL-bright zircon is related to Sveconorwegian reworking. Two CL-bright spots that plot along a recent lead loss line from 930 Ma were left out of the

Table 2 Regressions and weighted mean age calculations for zircon grains and populations. Errors are given at 2σ-level

Sample	Calculations (2σ errors)	MSWD	Comments
Hakefjorden Complex contact migmatite (HAS 96002)			
Grain 24. Two spots on detrital zircon.	Intercepts at 1634 + 44/–42 and 167 + 130/–150 Ma	–	Phanerozoic to recent lead loss supported. Upper intercept >1590 Ma which is the minimum age for the sedimentation (Åhäll et al. 1998).
Grain 25. Four spots on detrital zircon with overgrowth.	Intercepts at 1693 ± 380 and 447 ± 440 Ma	10	Three spots scatter near lower intercept. One spot close to upper intercept with $^{207}Pb/^{206}Pb$ age of 1680 ± 38 Ma.
Grain 38. Four spots on detrital zircon.	Intercepts at 2049 + 130/–93 and 904 + 130/150 Ma	1.3	Heat from the Hakefjorden Complex intrusion caused variable Pb-loss in Stora Le Marstrand Formation detrital zircons.
Grain 42. Five spots on detrital zircon.	Concordia age for three spots = 1612 ± 13 Ma Discordia intercepts at 1683 and 1131 with undefined errors	4.3 1.3	Some indication of lead loss due to the Hake-fjorden Complex intrusion. Regression anchored at 916 ± 16 Ma yields a 1608 ± 94 Ma (MSWD = 5.0) upper intercept.
Hakefjorden Complex back-vein (HAS 96003)			
Grain 21. Two spots on Xenocrystic zircon.	Intercepts at 1564 + 29/–28 and 319 + 29/–32 Ma	–	Lower intercept not geologically meaningful (multiple lead loss events?). Upper intercept is within error of 1590 Ma.
Grains 28, 51 and 57. Four spots on new zircon.	Concordia age = 916 ± 11 Ma	0.94	Spots concordant within 3% (excluding error overlap) were used. These new grains dates the intrusion of the Hakefjorden complex
Härsjön gneiss (OT 11)			
Grain 3. Two spots on detrital zircon.	Concordia age = 1639 ± 22 Ma	0.87	Age of provenance component.
Grain 66. Three spots on detrital zircon.	Mean $^{207}Pb/^{206}Pb$ age = 1675 ± 17 Ma	1.44	Normal regression not meaningful because of slight discordance, therefore the $^{207}Pb/^{206}Pb$ age is used. Age of provenance component.
Grain 83. Four spots on detrital zircon.	Concordia age = 1663 ± 16 Ma	0.90	Age of provenance component.
Grain 84. Three spots on detrital zircon.	Concordia age = 1639 ± 18 Ma	1.5	Age of provenance component.
Grain 85. Three spots on detrital zircon.	Concordia age = 1669 ± 20 Ma	2.9	"Student's *t*" 95% confidence level on error. Age of provenance component.
Olstorp back-vein (OT 16)		–	
Grains d7(.2), e2b, e6a and e7b. Four spots on new zircon.	Intercepts at 1620 ± 15 and –108 ± 403 Ma	0.04	Recent lead loss model supported. $^{207}Pb/^{206}Pb$ age = 1624 ± 6 Ma (MSWD = 0.16). Dates the intrusion of the Olstorp intrusion.
Borås tonalite (AA 9637)			
Grains 17, 21, 30 and 31. Five spots on igneous zircon.	Concordia age = 1674 ± 8 Ma	0.80	Only spots that are concordant within 0.5% (excluding error) were used. Dates the intrusion of the Borås intrusion.
Grain 17. Two spots on igneous and reworked igneous zircon.	Intercepts at 1665 ± 48 and 940 ± 270 Ma	–	Supports Sveconorwegian lead loss.
Grain 28. Two spots on igneous and reworked igneous zircon.	Intercepts at 1647 ± 100 and –279 ± 1300 Ma	–	Both spots near concordia.
Grain 30. Five spots on igneous and reworked igneous zircon.	Intercepts at 1671 ± 53 and 798 ± 88 Ma	5.1	Rough agreement with Sveconorwegian lead loss. One strongly reverse discordant spot excluded.
Grain 31. Four spots on igneous and reworked igneous zircon.	Intercepts at 1709 ± 28 and 937 ± 36 Ma	0.80	Young spots are reworked igneous zircon. Sveconorwegian lead loss is strongly supported.
Spots 4.1, 5.2, 17.1, 17.2, 21.1, 28.3, 30.1–30.3, 30.4c and 31.1–31.3.	Intercepts at 1683 + 15/–14 and 930 + 33/–34 Ma	0.84	A Sveconorwegian lead loss event is supported by zircon recrystallisation.

Table 3 Rare earth data for Hakefjorden Complex and Olstorp back-vein samples and the Borås tonalite

Element (ppm)	HAS96003 Hakefjorden Complex back-vein. Analysed with LA-ICPMS										OD16 Olstorp back-vein. Analysed with SHRIMP							AA9637 Borås tonalite: Analysed with SHRIMP							
	20Ax	20Bx	21Ax	21Bx	28An	28A2n	35Ax	61An	61Bn	61Cn	64Ax	D6Ax	D6Bx	D7Ax	D7Bn	D4bAx	E3bAx	E2bAn	E6AAn	17A	17B	17Cm	28A	30Am	30B	31Am	31B
P	726	680	395	442	n.a.	933	1845	n.a.	n.a.	1738	608	408	311	229	176	209	289	495	1494	237	313	49	421	45	488	57	607
La	16	14	22	117	3.6	66	70	0.9	3.9	3.0	43	19	25	0.1	3.0	0.03	0.1	10	12	0.3	1.8	0.3	1.1	1.9	0.8	1.5	0.6
Ce	99	103	135	593	25	125	365	22	26	33	225	391	267	32	33	17	33	131	99	83	84	6.5	84	11	55	10	117
Pr	14	14	25	116	4.9	24	85	2.4	4.8	5.3	38	65	23	0.6	1.4	0.1	0.8	7.8	9.0	1.4	3.3	0.4	2.2	2.2	1.1	2.2	2.4
Nd	80	78	138	653	43	94	509	38	49	52	220	373	103	9.1	6.8	1.3	10	44	44	18	33	4.0	22	21	9.2	18	29
Sm	40	37	79	319	48	48	388	60	59	69	119	84	47	14	2.6	1.9	14	18	17	25	35	3.4	30	13	12	10	41
Eu	14	13	25	101	9.4	8.1	89	6.6	8.5	8.8	36	39	12	2.2	0.9	0.5	2.7	5.3	3.1	2.4	4.8	0.6	3.1	3.8	1.6	2.0	3.7
Gd	95	97	149	468	190	185	780	295	250	302	217	131	68	42	7.0	8.1	38	44	37	84	78	7.3	87	20	28	12	135
Tb	31	32	37	96	63	57	202	96	84	100	56	37	13	13	2.5	3.0	12	14	8.8	25	22	1.4	27	3.3	9.4	1.8	45
Dy	404	390	368	812	789	715	1805	1095	947	1123	592	474	140	162	50	39	156	168	136	294	243	24	324	22	94	16	475
Ho	157	149	123	237	279	252	557	367	330	399	214	195	57	62	28	18	62	88	65	114	91	9.0	120	7.4	43	5.0	196
Er	769	727	555	990	1178	1058	2217	1614	1429	1672	986	763	254	242	149	77	242	397	300	449	354	42	468	31	172	27	662
Tm	196	180	122	228	258	235	442	325	313	345	248	172	59	47	41	17	52	83	73	89	68	10	89	7.3	37	6.9	147
Yb	1858	1695	1083	2113	2212	2082	3476	3145	2821	2735	2330	1821	594	447	463	191	454	958	690	742	576	146	768	98	350	82	1296
Lu	356	320	210	416	387	343	655	486	462	520	472	304	115	65	91	31	79	171	128	127	97	32	137	23	65	18	209
Lu/La_N	220	212	93	34	1021	50	90	5414	1128	1652	107	154	44	5689	292	9948	8451	165	103	4076	519	906	1199	117	754	116	3410
Eu/Eu*	0.68	0.66	0.70	0.80	0.30	0.26	0.49	0.15	0.21	0.19	0.69	1.14	0.65	0.28	0.62	0.38	0.36	0.58	0.38	0.16	0.28	0.38	0.19	0.72	0.27	0.57	0.15
Ce/Ce*	1.6	1.8	1.4	1.2	1.4	0.8	1.1	3.7	1.5	2.0	1.3	2.7	2.7	29.0	3.9	97.0	30.4	3.6	2.3	30.8	8.3	4.5	13.0	1.3	13.9	1.3	23.7

regression. In cases where both ancient and recent lead loss occurred, regressions may yield unreliable results with upper intercepts that are too old and lower intercepts that are too young (Mezger and Krogstad 1997).

Discussion

Origin of back-veins and tonalite

Relationships for the Hakefjorden Complex are rather straightforward. The intrusion is surrounded by a thermal aureole. Melting of metagreywacke rocks close to the norite intrusion produced an anatectic granite magma that back-intruded the crystallizing norite (Fig. 2). Thus the back-veining event was essentially coeval with the intrusion of the mafic magma. The anatectic melting of the banded metagreywacke country rock is easily seen in the field; the intrusion of a coeval granitic magma from depth is not required to explain the veining relationships. The relationships at Olstorp are similar, although complicated by post-intrusive deformation of some contacts and by the small age differences between country rock detrital zircon and that of the mafic intrusion.

Åberg (1997) postulated and modelled a mixing origin for the Borås tonalite, because field relationships show that the tonalite formed by mixing between a gabbro component and the granitoid country rock. A least squares model based on major element analyses suggests the tonalite is a mix of 66% hornblende gabbro and 44% granitoid gneiss. The fit for all major elements is better than 1% relative but the trace element mixing using these proportions was more variable. LILE and compatible trace elements fit within 10% while P, Zr, Hf and Y in the model are 20–70% higher than the actual tonalite and the REE show a progressively worse fit towards the heavy REE. A possible explanation for these geochemical discrepancies is that refractory minerals such as zircon remained in the residue after melt extraction. This interpretation is further supported by the observation of centimeter-thick zircon-rich biotite selvages between mafic and felsic rock at Olstorp and by the lack of xenocrystic zircon in the sampled tonalite. Despite the shortcomings of simple modelling for some trace elements the field relationships and major element results are compelling evidence that the tonalite is the result of mixing between the intrusive gabbro and country rock. Even if this interpretation is completely incorrect, the tonalite clearly belongs to the same intrusive event as the mafic rocks and its zircon date is thus valid for the whole intrusion.

Development of new magmatic zircon in anatectites

In all three examples studied, the melting of country rock led to the formation of new zircon that can be used for ion probe dating. Overgrowths on old zircon are, in

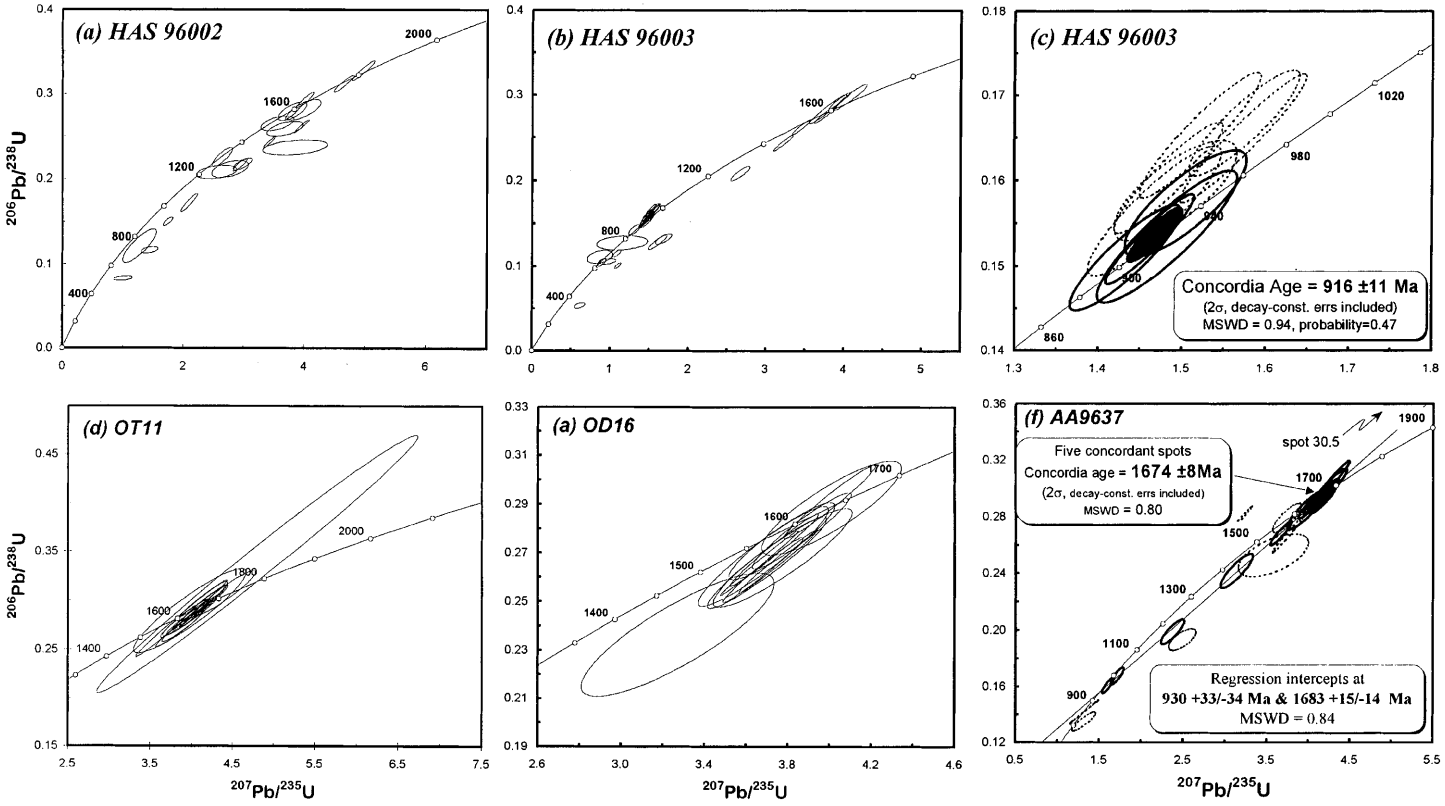


Fig. 5 Wetherill concordia diagrams for **a** all analysed grains from Hakefjorden Complex contact migmatite (HAS96002), **b** all analysed grains from Hakefjorden Complex back-vein granite (HAS96003), **c** new Hakefjorden Complex back-vein grains with calculated 916 ± 11 Ma ellipse in black (excluded discordant grains are shown with dashed lines), **d** Härsjön supracrystal from Olstorp (OT11), **e** Olstorp back-vein sample (OD16) and **f** Borås tonalite (AA9637)

all but a few cases, too thin to be dated with the ion-probe. However, new grains also formed in most samples. These tend to have relatively high aspect ratios, with high Th and U, CL-dark images and weak zonation. REE-profiles of both overgrowths and new grains are consistent and usually distinct from those of xenocrysts, which often have CL-bright, zoned images.

The non-destructive nature and high spatial resolution of ion-probes is very useful for U–Pb dating complex zircon and can also be used for REE-analyses of dated spots. REE-analyses can also be done after dating, using laser ablation ICPMS but this tends to destroy the zircon. In cases like the Hakefjorden Complex, with large age differences, new zircon is easily identified by its age and can provide much more precise ages of mafic–ultramafic intrusions than the Sm–Nd method. In addition, information about xenocrysts from the country rocks is gained.

In cases like the Olstorp intrusion where small age differences between the intrusion and xenocrysts are a problem, REE, Th and U analyses can be used in conjunction with CL images to characterize different zircon types. This approach is also useful in the case of metamorphic reworking as demonstrated for the Borås tonalite. The low-Th/U ratio, lower heavy REE concentrations and reduction of Ce- and Eu-anomalies are diagnostic in this example and in several other metamorphic zircons we have studied.

Lead loss in anatectic and metamorphic environments

The retention or loss of lead from zircon in the anatectic magmatic environments described here is critical to interpretation of the age data. According to Mezger and Krogstad (1997) crystalline zircon is unlikely to lose Pb by self-diffusion between 600 and 1000 °C, and the back-veining and mixing events probably occurred within that temperature range. However, metamict or partly radiation-damaged zircon in the xenocrystic population could still be prone to lead loss that could lead to partial or complete resetting of a xenocrystic grain. It might then date the intrusion if completely reset or it could plot along a Pb-loss line between the age of origin and intrusion. Zircon often shows strong zonation in U-concentration reflected by variations in CL-brightness, probably related to high zircon/liquid partition coefficients and slow U diffusion rates in the magma. Ancient Pb-loss as a function of U-concentration in multiply analysed grains can be investigated by plotting $^{207}Pb/^{206}Pb$ age against U-concentration as in Fig. 7 for Hakefjorden Complex xenocrysts. Four of the six grains show distinct trends towards lower ages with increasing U-concentration. Grain 96002:38 shows an opposite trend because of an old, high-U core spot. The inverse correlation for four grains, and with eight ages being

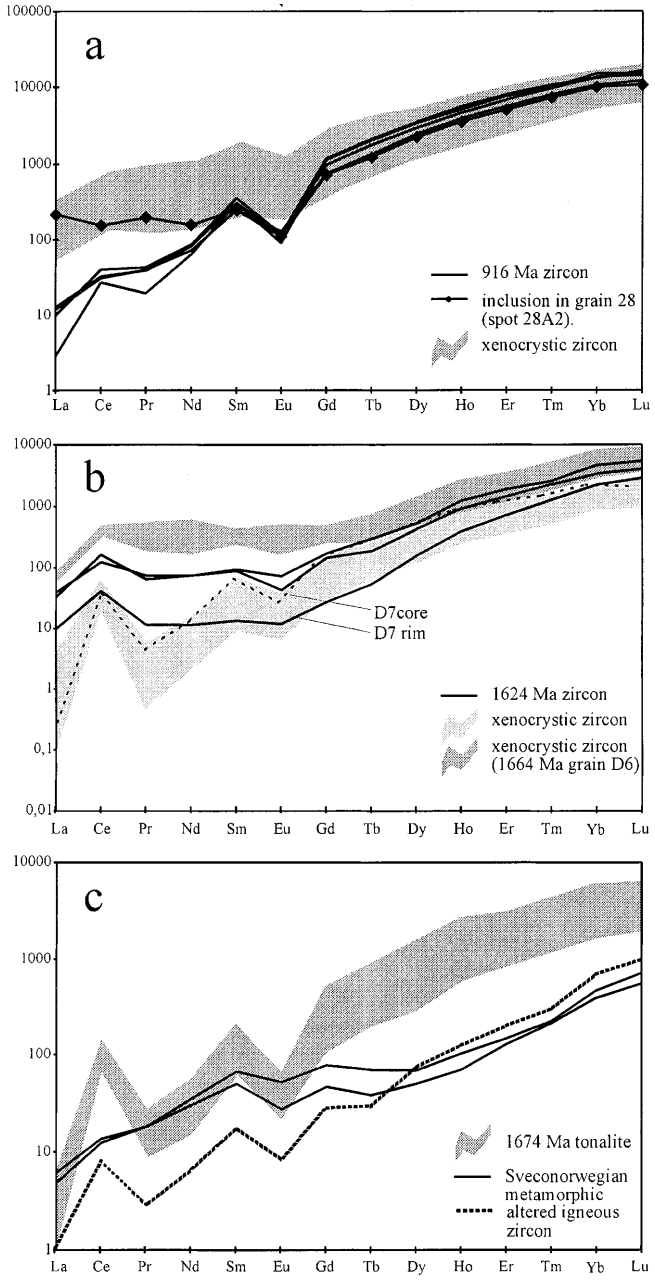


Fig. 6 Chondrite normalized (Boynton 1984) REE-profiles for **a** Hakefjorden Complex back-vein sample (HAS96003), **b** Olstorp back-vein sample (OD16) and **c** Borås tonalite (AA9637)

lower than the minimum age of 1587 ± 3 Ma deduced for the country rocks by Connelly and Åhäll (1996) shows that ancient Pb-loss did occur and, according to the regressions for grains 38 and 42 in Table 2, this happened close to the time of Hakefjorden Complex emplacement. Åhäll et al. (1998) did not find any evidence for Sveconorwegian Pb-loss in the regionally metamorphosed Stora Le-Marstrand metagreywackes reinforcing our interpretation that the Pb-loss is related to the Hakefjorden Complex. We envisage that Pb-loss occurred from U-rich zones via fission tracks and the cracks that are seen in the CL images.

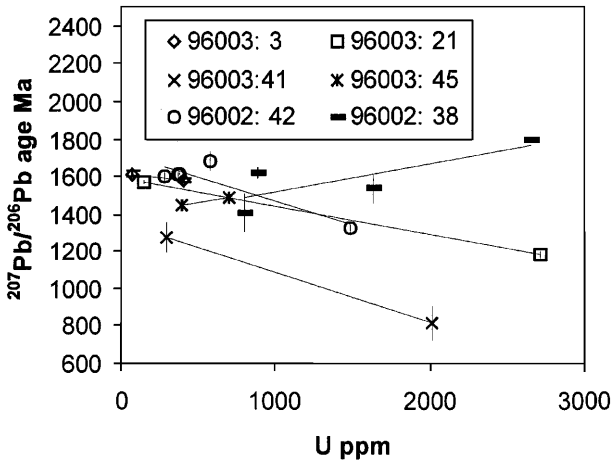


Fig. 7 U-concentration against $^{207}Pb/^{206}Pb$-age for Hakefjorden Complex xenocrysts. The diagram demonstrates increasing ancient Pb-loss as a function of increasing U-concentration in four grains. High-U zircon is more likely to suffer crystal structure damage and therefore Pb-loss. The U-rich core that did not suffer Pb-loss may have been shielded or did not suffer crystal damage in spite of its high U-concentration

Pb-loss associated with the intrusion at Olstorp is difficult to identify because of the small age differences involved. All spots on zircon from the intruded paragneiss plot on or very near the concordia and nothing seems to suggest Pb-loss. The xenocrysts in the back-vein sample were only analysed with one spot each and this precludes any analysis of Pb-loss. However, when the $^{207}Pb/^{206}Pb$ minimum ages are plotted on relative probability histograms, they show two distinct peaks at ~1635 and ~1665 Ma, similar to the ~1640 and ~1665 Ma peaks for the Härsjön paragneiss. One concordant and three discordant back-vein grains conforms with recent lead loss which in conjunction with the data for the country rock paragneiss indicate no Sveconorwegian metamorphic Pb-loss.

Upper amphibolite metamorphism at 700 °C did affect the 1674 ± 8 Ma igneous zircons in the Borås tonalite. The discordia obtained by including discordant analyses of the grains defining the 1674 ± 8 Ma age has a lower intercept of 930 + 33/–34 Ma indicating to Sveconorwegian metamorphism. The age is rather young compared with other zircon dates around 960–970 Ma but compares with 920–950 Ma titanite ages (e.g., Connelly et al. 1996; Möller and Söderlund 1997; Wang et al. 1998, and our unpublished data). This may be explained in terms of recent Pb-loss, that is noted for two spots, so the 930 + 33/–34 Ma age should be considered a minimum age. In this case we envisage that the reworking of this U-rich igneous zircon along fractures, embayments and via fission tracks (Fig. 4e). This lowered Th/U due to predominant Th-loss, P was also lost and REE exchanged, eliminating the Ce-anomaly (Fig. 6c). These changes might be explained in terms of monazite solid solution in magmatic zircon that was removed when the zircon re-equilibrated with metamorphic fluids at lower temperature. Monazite

exsolution could account for the observed changes in zircon composition. From textural evidence in the zircon grains it seems that such a reaction could be fluid driven rather than temperature driven since these changes occur along fractures and embayments.

Conclusions

New zircon forms during anatectic melting of intruded country rock. It usually forms new elongate grains as well as overgrowths on zircon xenocrysts. By careful study of CL images and selection of ion probe spots in zircon grains from anatectic melts and back-veins mafic–ultramafic intrusions can be precisely and accurately dated. Xenocrysts from the country rock can also be dated. They are distinguished from new magmatic zircon using CL images and microanalysis for REE, Th and U. Xenocrysts should also be tested for Pb-loss that may occur in U-rich zircon because of the heat of the intrusion. Metamorphic events long after the time of intrusion can also cause Pb loss from U-rich zircon. This is driven by metamorphic fluids, that can effectively remove the monazite component from magmatic zircon, resulting in the preferential loss of Th, P and Ce.

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